

Level 4

Additional Resources

The Endocrine System

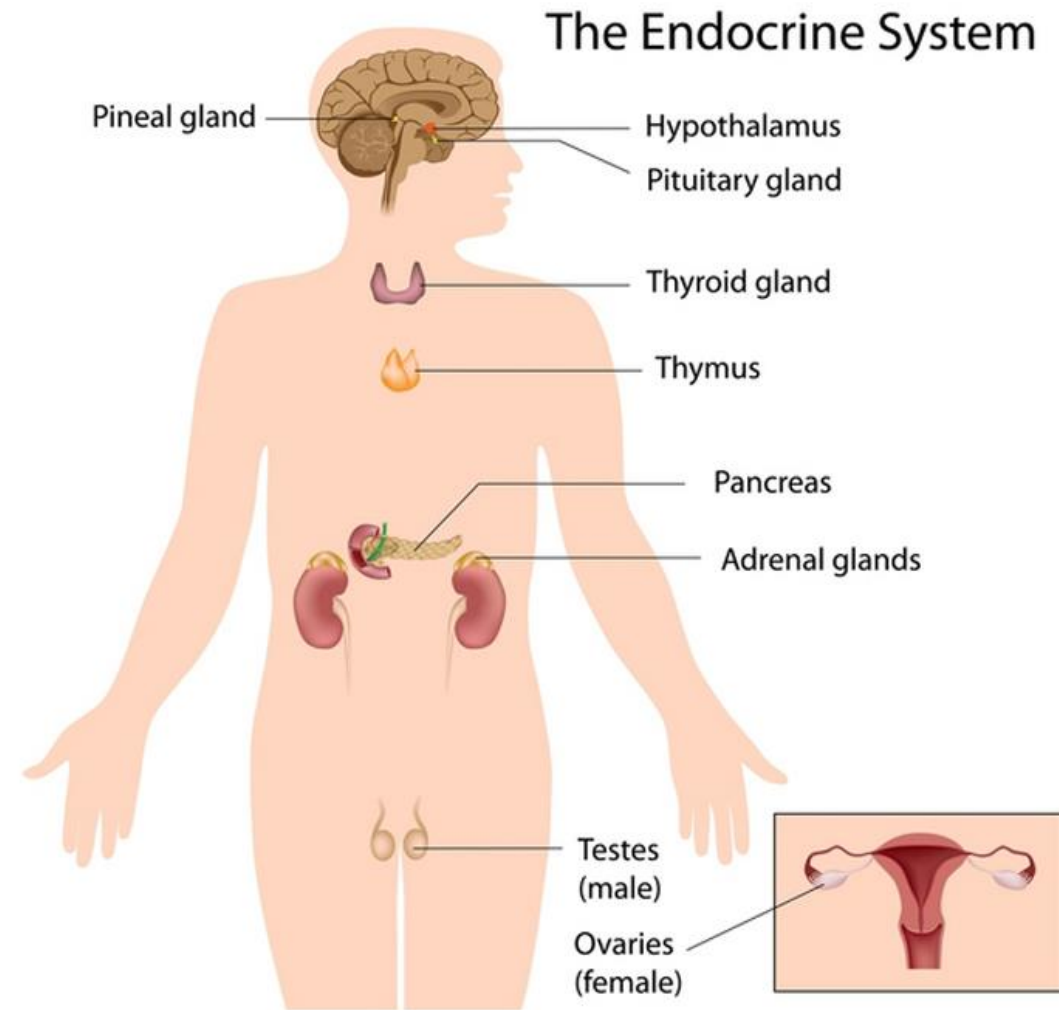
- Made up of a network of glands
- Glands are small organs which produce, store and release hormones

Main Functions of the Endocrine System

- Metabolism
- Emotions and Mood
- Growth and Development
- Fertility and sexual function
- Sleep
- Response to stimuli

Endocrine Glands

- Hypothalamus (base of the brain)
- Pineal
- Pituitary
- Thyroid and Parathyroid
- Thymus
- Adrenal Gland
- Pancreas
- Ovary



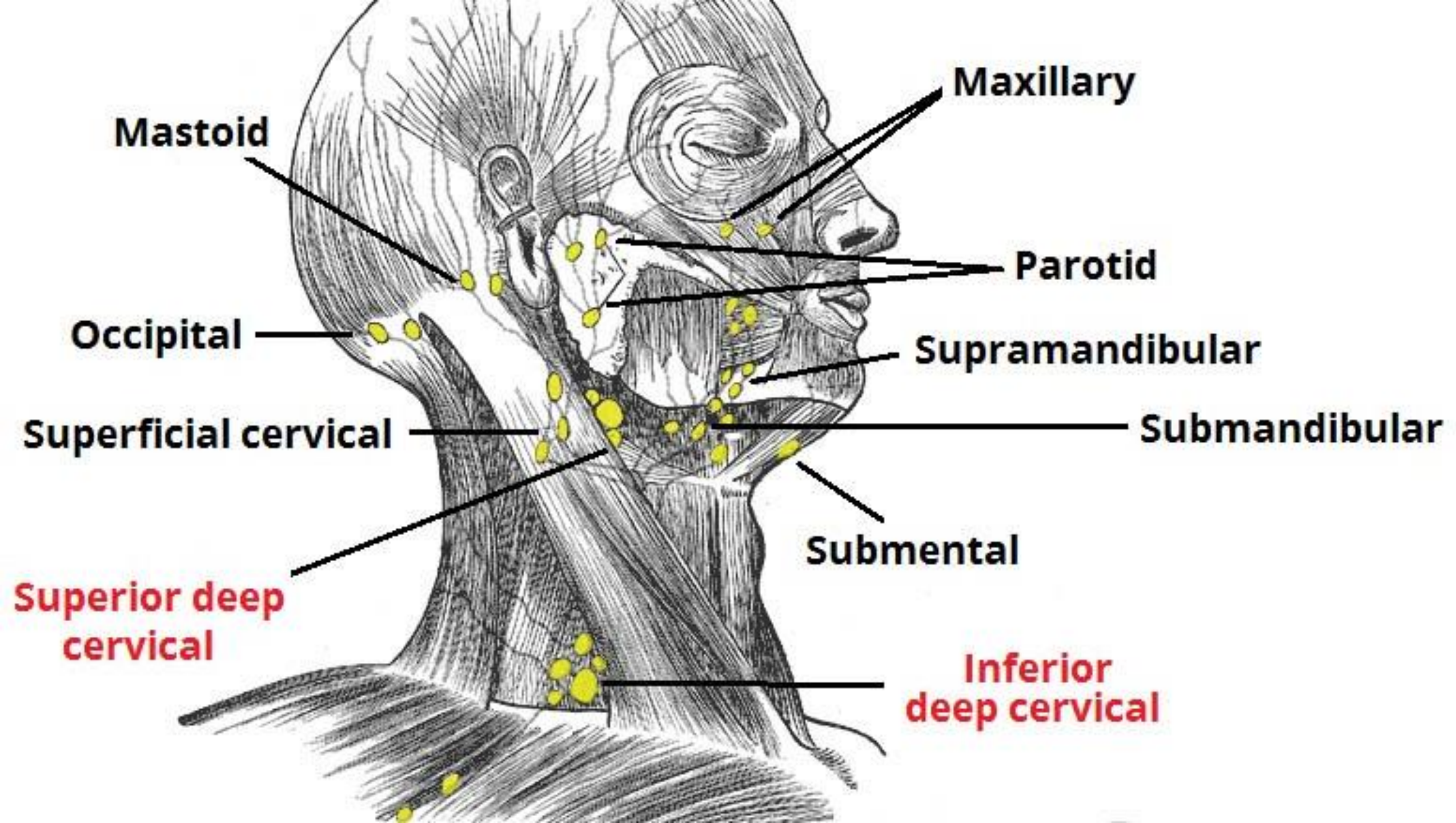
Effects on the Skin

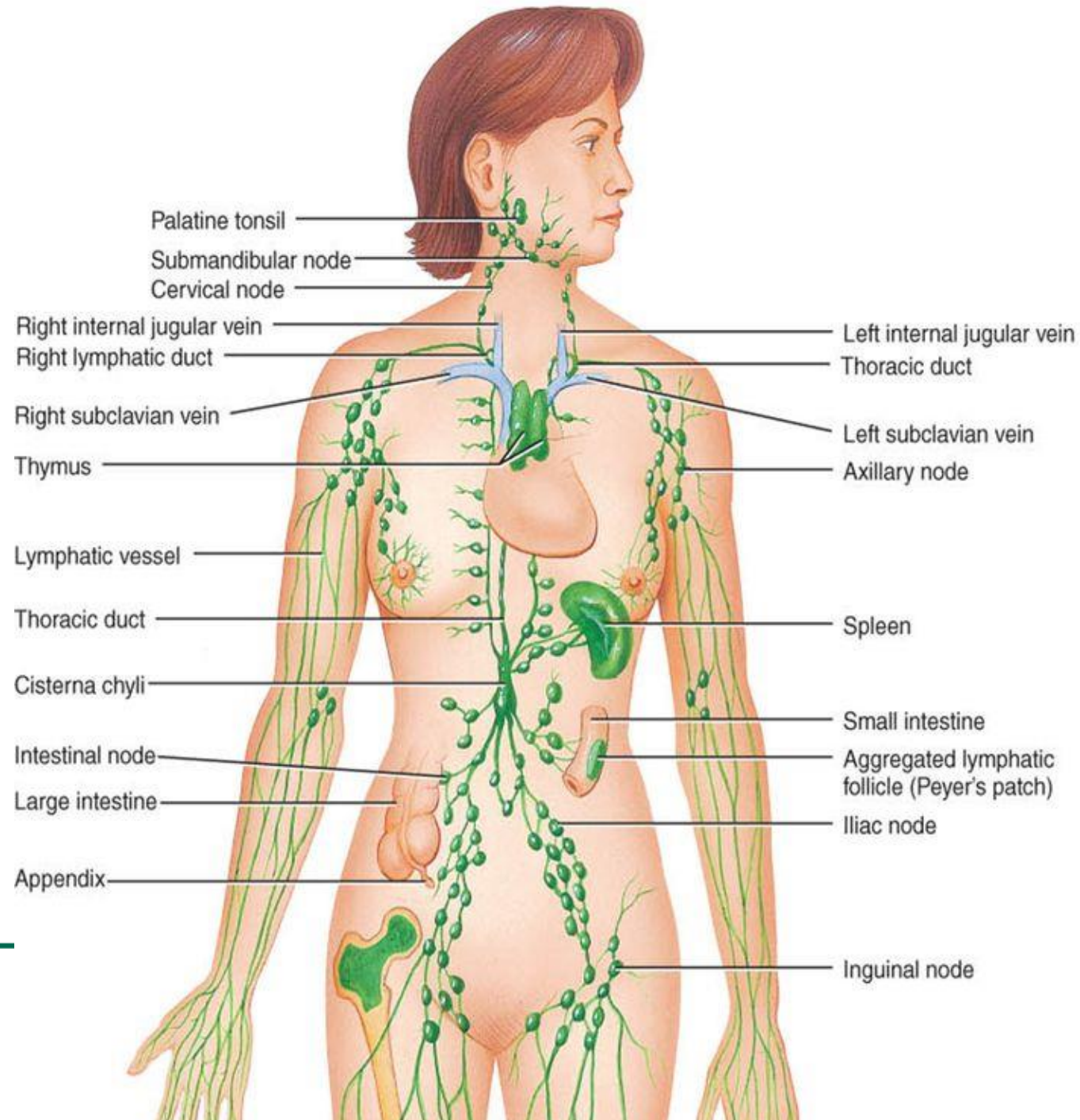
- Hormones affect the oil production within the skin. This will increase the likelihood of the follicles being congested therefore resulting in inflammation and pustules

Effects on the Hair

- Hormones are responsible for the hair growth
- An endocrine dysfunction can affect the hair growth cycle and thickness
- Estrogen (female hormones) can contribute to slowing down the growth by prolonging the anagen phase
- Androgenic (male hormones) can accelerate the hair growth

The Lymphatic Syestem



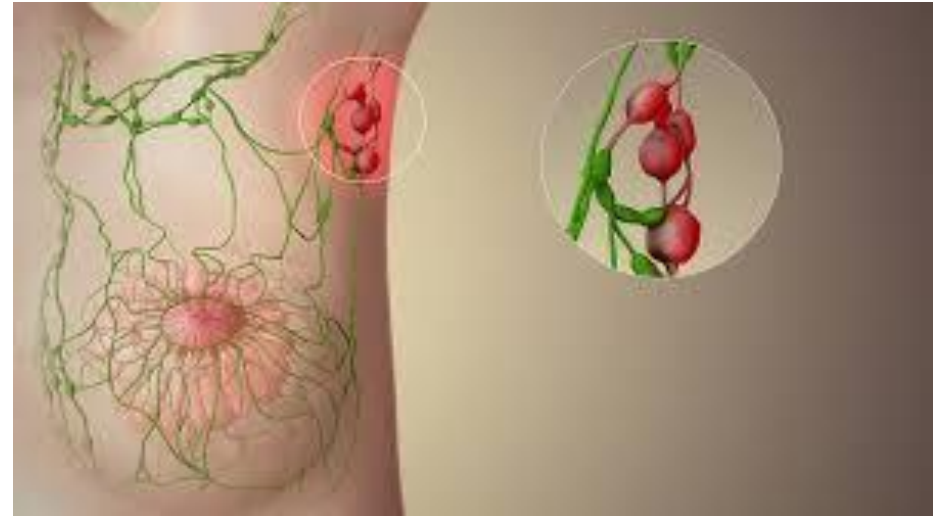
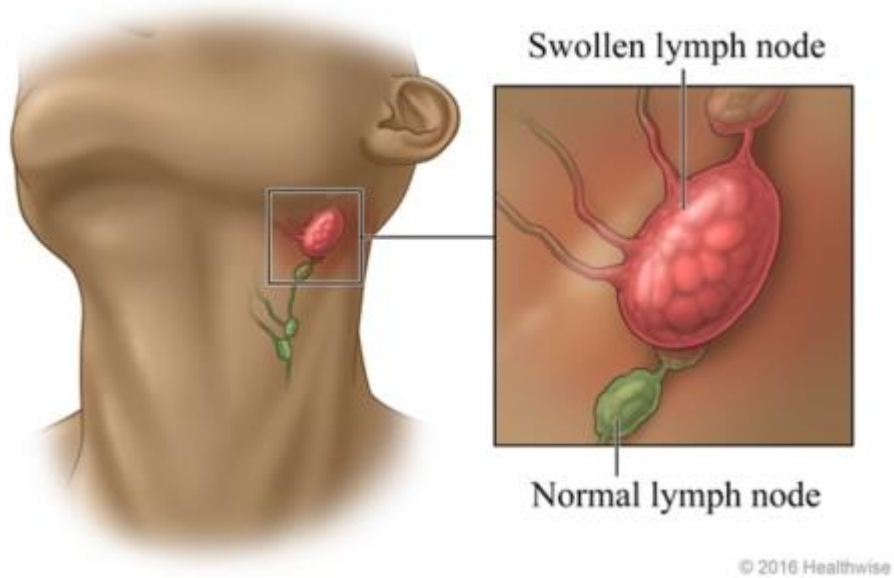


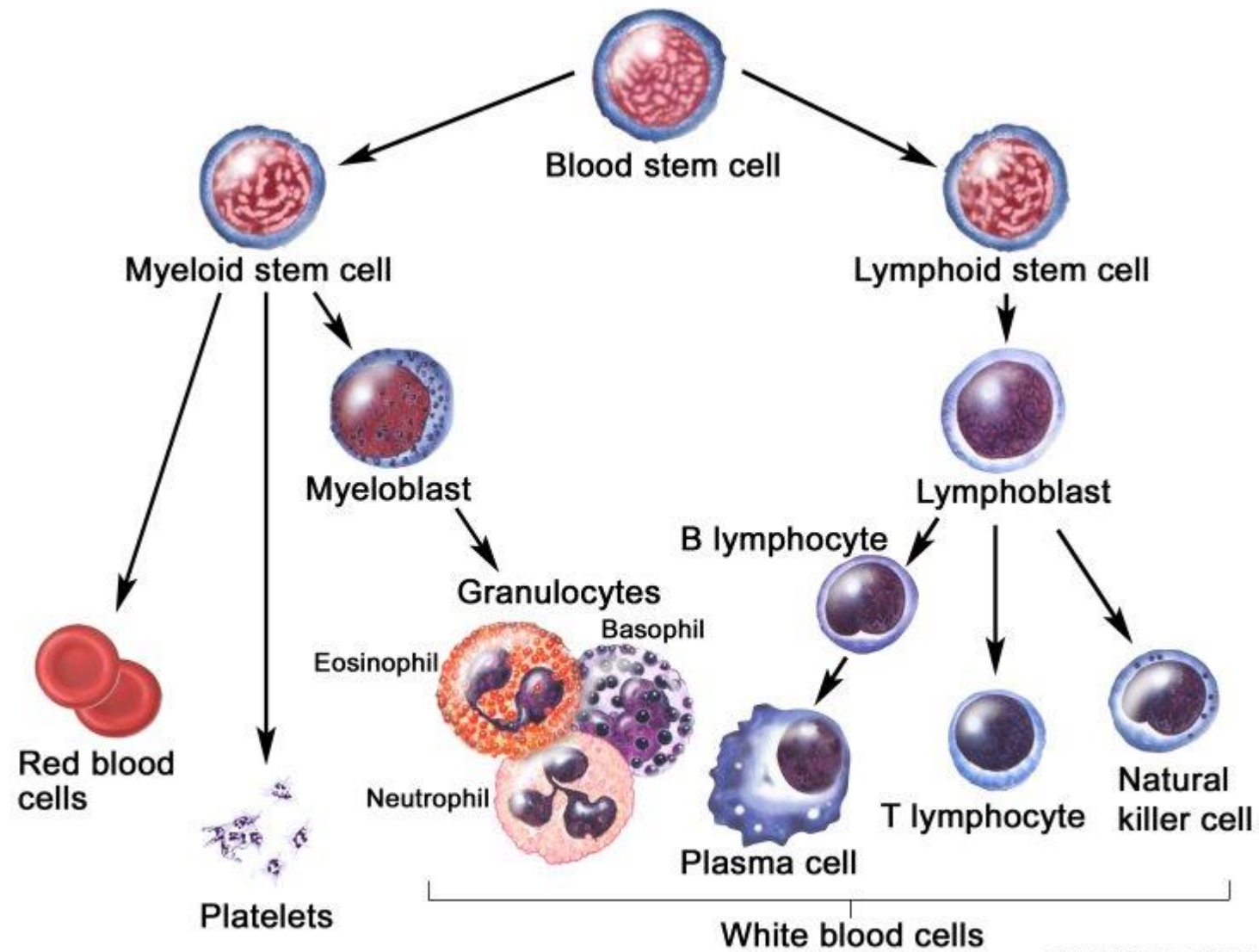
Lymphatic System



What is Lymphatic fluid and what's in it?

- Lymph fluid normally flows through a network of lymph vessels. These lymph vessels connect to a group of lymph nodes. The nodes act as a filter. They trap or destroy anything harmful that the body does not need
- Inside the lymph nodes are white blood cells, also called lymphocytes. These white blood cells attack and break down bacteria, viruses, damaged cells or cancer cells.
- The lymph fluid carries the waste products and destroyed bacteria back into the bloodstream. The liver or kidneys then remove these from the blood. The body passes them out with other body waste, through bowel movements.
- Lymph nodes sometimes trap bacteria or viruses that they cannot destroy straight away. For example, they may do this when you have an infection. When the lymph nodes are fighting the infection, they often swell and become sore to touch.





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Lymphocyte

- A lymphocyte is a type of white blood cell in the immune system of most vertebrates.
- Lymphocytes include natural killer cells, T cells, and B cells. They are the main type of cell found in lymph, which prompted the name "lymphocyte".
- Lymphocytes make up between 18% and 42% of circulating white blood cells.
- Both originate from stem cells in the bone marrow and are initially similar in appearance.
- Some lymphocytes migrate to the thymus, where they mature into T cells; others remain in the bone marrow, in humans—they develop into B cells.
- Most lymphocytes are short-lived, with an average Life span of a week to a few months, but a few live for years, providing a pool of long-lived T and B cells. These cells account for immunologic “memory” a more rapid, vigorous response to a second encounter with the same antigen.

